

INTERNSHIP REPORT

Software Development Internship — Code UpScale

Organization	Code UpScale
Department	Software Development
Role	Software Development Internee
Supervised By	Usama Mushtaq
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Roll Number	BSSE 23095
Duration	July 20, 2026 – September 1, 2026 (6 weeks)

Overview of the Internship

I completed a six-week Software Development internship at Code UpScale, working within the Software Development department alongside the engineering team. The internship followed a hands-on, project-based structure: I was assigned a live full-stack application to build and extend, with daily progress updates and iterative code review from senior team members. Work was organized and tracked through the team's internal communication channel, where tasks, resources, and feedback were shared throughout the internship.

Project: Full-Stack E-Commerce Platform

The core deliverable was a full-stack e-commerce web application for an online jewelry store, covering both the customer-facing storefront and an internal admin panel. The project was built on the MERN stack (MongoDB, Express.js, React, Node.js), split into a separate frontend and backend codebase. The application is deployed and publicly accessible at <http://areesha-co-env.eba-dyurxx2e.ap-south-1.elasticbeanstalk.com/>.

Technical Contributions — Frontend

- Built the React 19 storefront using functional components, React Router for navigation, and Tailwind CSS for styling.
- Developed core shopper flows: product catalog, category and collection pages, product detail pages, cart, and checkout (Place Order).
- Implemented a global shopping-cart context (React Context API) to manage cart state, product data, and currency formatting across pages.
- Built admin-facing screens for adding, editing, listing, and removing products (with multi-image upload for new products), backed by dedicated admin authentication separate from customer accounts.
- Added product search and category filtering, and integrated a curated product dataset (JSON) used to seed and display the catalog.

Technical Contributions — Backend

- Developed REST APIs using Express.js for user authentication, product management, and order processing, with MongoDB and Mongoose for data persistence.
- Implemented JWT-based authentication and role-gated middleware to separate customer and admin access.

- Integrated Cloudinary for product image uploads, using Multer to handle multipart form data before forwarding files to cloud storage.
- Implemented order creation, retrieval, and status-tracking endpoints tied to authenticated users, with Cash on Delivery as the available payment method and the checkout flow structured to support online payment integration in the future.
- Wrote a database seed script to populate the product catalog from structured JSON data for testing and demonstration.

Skills and Tools Used

React, JavaScript (ES6+), Tailwind CSS, Node.js, Express.js, MongoDB/Mongoose, JWT authentication, Multer, Cloudinary, Axios, Git, REST API design, and AWS Elastic Beanstalk for deployment. The internship also involved reading and adapting to an existing codebase, debugging across the frontend/backend boundary, and incorporating feedback from code reviews.

Results Gained

- Developed and deployed a working full-stack e-commerce application to AWS Elastic Beanstalk, covering database integration, REST APIs, authentication, product management, and the customer-facing interface.
- Built end-to-end shopper journeys — browse, search, cart, and place order — and admin tools for product management.
- Gained practical, project-tested experience with authentication, third-party API integration (Cloudinary), cloud deployment (AWS Elastic Beanstalk), and full-stack debugging.
- Developed stronger habits around structured task tracking, daily reporting, and responding to code review feedback in a team setting.

Conclusion

This internship gave me direct, practical experience building a production-style full-stack application rather than isolated exercises. Working across the frontend and backend of a real e-commerce project deepened my understanding of how modern web applications are structured end-to-end, and improved my confidence in React, Node.js/Express, and MongoDB. I'm grateful to the team at Code UpScale for their mentorship and for the opportunity to work on a meaningful project throughout the internship.